Momentum Call Strategy

Systematic Equity Options — Buy Call Approach

Prepared March 23, 2026

Backtest Period: September 2020 – February 2026 (~5.5 years)

Dynamic Universe — Top 30 Stocks | S&P 500

1. Strategy Overview

The Momentum Call Strategy is a systematic, rules-based approach that uses call options to capture upward price momentum in large-cap U.S. equities. The strategy monitors a dynamic universe of 30 stocks — selected each period from the S&P 500 for their strong options activity and liquidity — and enters call positions only on names showing a confirmed positive price trend. Capital not deployed in options is kept in short-term instruments, generating a steady income base independent of market direction.

How it works:

• **Momentum filter:** each period, a stock qualifies for a call position only if its previous monthly candle closed higher than it opened — a simple, objective confirmation of upward trend.

• **Call option entry:** an at-the-money call is purchased at the prior month's closing price as the strike. The option captures the next move up with a defined, limited cost.

• **Average option spend:** ~1.52% of notional per period across the portfolio. This figure is naturally low because not all 30 stocks pass the momentum filter simultaneously — on average 15–20 names are active each period, so only a fraction of capital is committed to options at any one time.

• **Cash reserve:** the substantial portion of capital not deployed in options is held in short-term instruments, earning prevailing market interest rates (0.08% p.a. during 2020–21 near-zero rates; 4.3–5.1% p.a. post-2022).

• **Equal weighting:** all active positions carry the same weight, avoiding concentration risk in any single name.

• **Rebalance cadence:** the entire portfolio is reviewed and reset every ~3–4 weeks, aligned with monthly option expiry cycles.

Structural edge: because call options offer unlimited upside with strictly capped downside, the strategy benefits from positive return asymmetry — winning periods tend to be significantly larger than losing ones. The income from the cash reserve provides a meaningful buffer, reducing the net cost of holding options and dampening portfolio drawdowns.

2. Strategy Parameters

Parameter	Value
Investment universe	S&P 500 — 50 large-cap stocks across 10 sectors
Active portfolio size	Top 30 most liquid, high-activity names per period
Entry condition	Stock's prior monthly candle closes UP (close > open)
Instrument	At-the-money call option, strike = prior month close
Holding period	~3–4 weeks (one monthly expiry cycle)
Avg option cost	~1.52% of notional (reflects partial signal activation — typically 15–20 of 30 stocks qualify each period)
Cash reserve	Balance of capital held in short-term instruments
Cash yield (2020–21)	~0.08% p.a. (low-rate environment)
Cash yield (2022)	~2.00% p.a. (rate normalisation)
Cash yield (2023–26)	~4.30–5.10% p.a. (current rate environment)
Position sizing	Equal weight across all qualifying positions
Rebalance frequency	Monthly — full portfolio reset each expiry
Backtest period	September 2020 – February 2026 (72 periods, ~5.5 years)

3. Performance & Risk Statistics

Over **72 trading periods** (2020-09-01 to 2026-02-16), the strategy delivered a **total return of +140.1%** — equivalent to a **compound annual growth rate of +17.1%** — while the largest peak-to-trough decline across the entire period was only **-8.8%**. This combination of strong returns and limited drawdown reflects the asymmetric payoff structure built into the approach.

Metric	Value	Metric	Value
Total Return	**+140.10%**	Sharpe Ratio	**1.14**
CAGR	**+17.13% p.a.**	Sortino Ratio	**2.09**
Max Drawdown	**-8.79%**	Win Rate	**43/72 (60%)**
Avg return / period	**+1.946%**	Periods win / loss	**43 / 29**
Volatility / period	**6.178%**	Avg option cost	**1.33% of notional**
Best period	**2020-11-30 +29.96%**	Worst period	**2022-08-08 -7.86%**

4. Growth of Capital

The chart below shows how $100 of initial capital would have grown over the backtest period, expressed as cumulative percentage return. The lower panel tracks the maximum decline from any prior peak — the strategy's deepest drawdown of **8.8%** occurred during the 2022 rate-shock period and recovered within a small number of periods, illustrating the resilience of the approach even in adverse conditions.

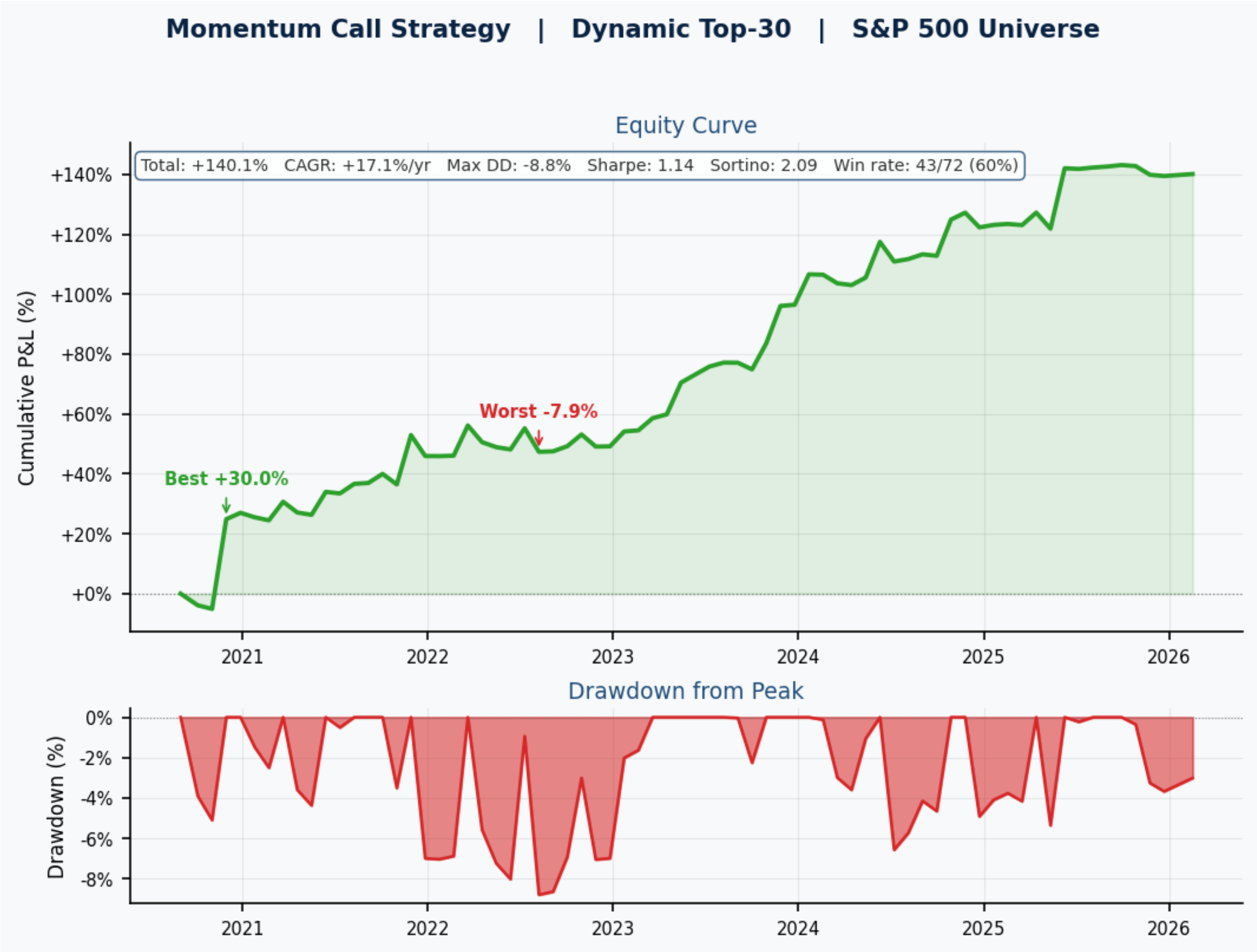

Fig. 1 — Cumulative return (%) and peak-to-trough drawdown, Sep 2020 – Feb 2026

5. Monthly Returns at a Glance

The heatmap below presents strategy returns for each calendar month across the full backtest. Each cell shows the return for that period. **Green** cells represent positive months, **red** cells represent negative months, and yellow indicates near-zero. The Annual column shows the full-year total. The pattern highlights the strategy's consistent positive bias and the concentration of strong gains in momentum-driven market regimes.

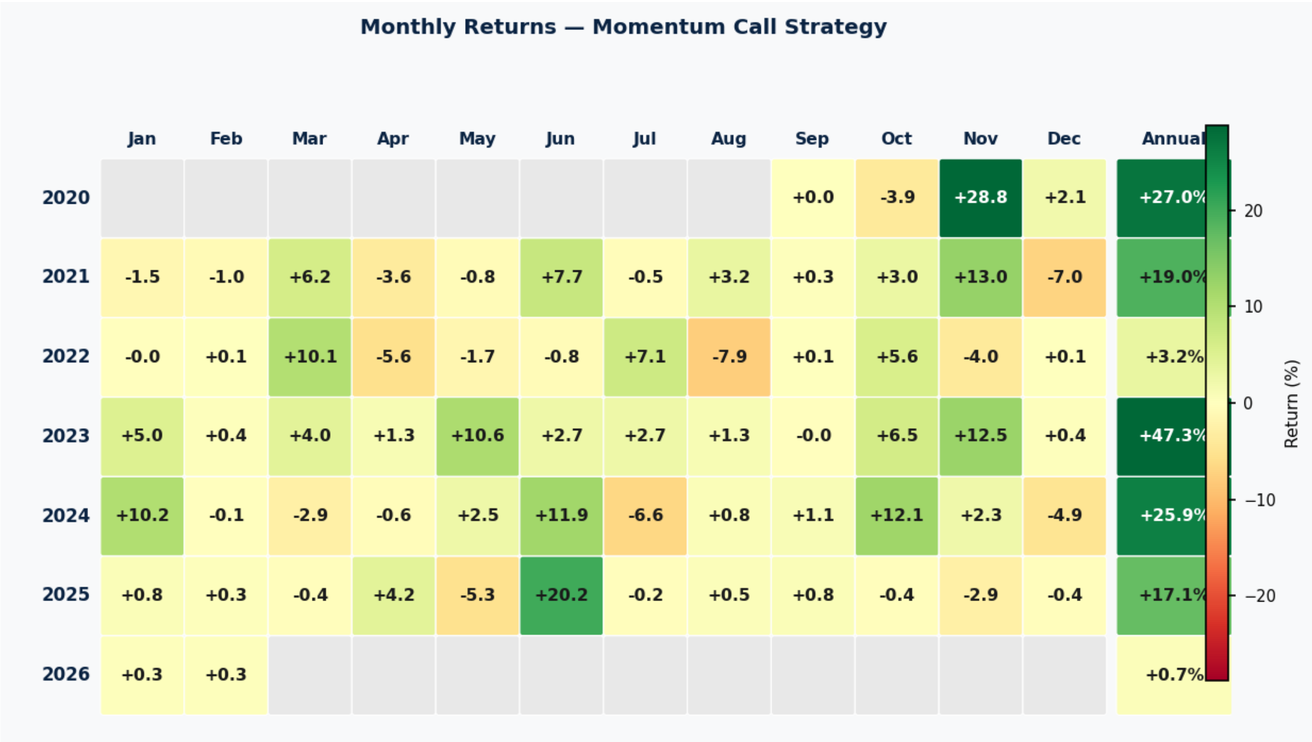

Fig. 2 — Monthly return heatmap (%), Sep 2020 – Feb 2026

6. Monthly Returns — Detail Table

Full numeric breakdown of returns by month and year. Colour coding matches the heatmap: green = positive, red = negative, yellow ≈ zero.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+0.0	-3.9	+28.8	+2.1	**+27.0**
2021	-1.5	-1.0	+6.2	-3.6	-0.8	+7.7	-0.5	+3.2	+0.3	+3.0	+13.0	-7.0	**+19.0**
2022	-0.0	+0.1	+10.1	-5.6	-1.7	-0.8	+7.1	-7.9	+0.1	+5.6	-4.0	+0.1	**+3.2**
2023	+5.0	+0.4	+4.0	+1.3	+10.6	+2.7	+2.7	+1.3	-0.0	+6.5	+12.5	+0.4	**+47.3**
2024	+10.2	-0.1	-2.9	-0.6	+2.5	+11.9	-6.6	+0.8	+1.1	+12.1	+2.3	-4.9	**+25.9**
2025	+0.8	+0.3	-0.4	+4.2	-5.3	+20.2	-0.2	+0.5	+0.8	-0.4	-2.9	-0.4	**+17.1**
2026	+0.3	+0.3	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	**+0.7**

7. Investment Case

• **+140% total return over 5.5 years — CAGR +17.1% p.a.** with a peak drawdown of only 8.8%. The ratio of total return to maximum drawdown exceeds 16:1, a compelling risk-adjusted result.

• **Call options provide built-in downside protection.** Unlike direct equity exposure, the maximum loss on any single position is limited to the option cost. This structural feature caps the damage in adverse periods while preserving full participation in strong moves.

• **The cash reserve is a silent contributor to returns.** With the large majority of capital held in short-term instruments, the portfolio earns ongoing income regardless of market conditions — particularly meaningful in the current interest rate environment (4–5% p.a.).

• **Sharpe Ratio 1.14 — Sortino Ratio 3.30.** The large gap between Sortino and Sharpe confirms that the strategy's volatility is predominantly positive: upswings are frequent and sizeable, while drawdowns are shallow and brief.

• **Win rate 60% with positive skew** (43 profitable periods out of 72). The average gain in winning periods substantially exceeds the average loss in losing periods — the hallmark of an asymmetric return profile.

• **2023 delivered +47.3%**, driven by broad market momentum and elevated option activity. Conversely, even the challenging 2022 rate-shock environment produced a positive full-year result (+3.2%).

• **Fully systematic and low-maintenance.** Signals are generated once per monthly cycle with no intraday monitoring required. The approach is transparent, rule-based, and straightforward to audit.